A10-STMS: A Reduced-Surrogate Audit of Mission-Variable-Preserving Hybrid Thermal Control for Spaceborne Industrial Systems under Radiative, Storage, Power, Delay, and Actuator Constraints

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Abstract

This paper introduces A10-STMS, a mission-variable-preserving hybrid thermal-control framework for spaceborne industrial systems operating under radiative, storage, power, sensing-delay, and actuator-authority constraints. The objective is not to propose a new heat-transfer mechanism, spacecraft hardware design, or flight-ready thermal-control system. Instead, A10-STMS is formulated as a closed nondimensional surrogate for testing whether separated-barrier hybrid control can preserve thermal safety and mission throughput in regimes where passive radiation, simple pump control, or pump–radiator control alone are insufficient.

The surrogate state consists of core temperature, cold-plate temperature, radiator temperature, storage margin, power margin, thermal-gradient risk, and mission throughput. The control channels are pump authority, thermal-path valve steering, radiator/emissivity authority, storage coupling, and load shaping. A staged numerical audit was conducted. Initial tests showed damage-limiting behavior but not mission feasibility under insufficient thermal resources. A resource-frontier sweep then identified a high-resource regime in which A10-Hybrid-v2 achieved mission-feasible separation from a Pump+Radiator baseline. Subsequent ablation, out-of-sample, delay, and actuator-degradation tests showed that routing, storage, load shaping, and authority-aware retuning all contribute to the final result.

The final conservative controller, A10-v4-safe, preserved manageable-stress feasibility under baseline, delayed-sensing, moderate actuator-degradation, and out-of-sample conditions while remaining separated from the Pump+Radiator baseline in failure rate and tail-risk metrics. The final locked numerical status is `PASS-PHASE2E-DELAY-DEGRADE-ROBUSTNESS`. Harsh combined stress remains unsolved and is explicitly excluded from the mission-feasible claim. The results support A10-STMS as a feasibility-diagnostic reduced-control architecture, not as a validated spacecraft thermal-control technology.

Keywords: spacecraft thermal control; reduced surrogate; mission-variable preservation; tail risk; CVaR; control barrier; delay compensation; actuator degradation; hybrid thermal management; A10 structured prior.

1 Introduction

Spaceborne industrial systems such as high-power communications payloads, orbital data-processing units, directed-energy subsystems, in-space manufacturing devices, electric-propulsion power electronics, and precision optical instruments impose increasingly coupled thermal requirements. Their thermal-control problem is not merely to reduce average temperature. A mission system can fail even when its average thermal load is acceptable if a local core, cold plate, radiator, storage buffer, power margin, or structural-gradient constraint is violated for a sufficiently short but critical interval.

Conventional spacecraft thermal-control engineering emphasizes conduction, radiation, surface coatings, heaters, louvers, heat pipes, pumped loops, thermal straps, storage, and mission-specific thermal margins [1–3]. These technologies are not replaced in this work. A10-STMS instead asks a narrower control-theoretic question: given finite radiator authority, heat-transport authority, storage capacity, and power margin, can a structured hybrid controller allocate cooling effort, thermal routing, storage, and load shaping so that a mission variable remains viable?

This question is related to safe-set and barrier-function control [5], model-predictive control [6], delayed-loop compensation [7], and tail-risk evaluation using Conditional Value-at-Risk (CVaR) [4]. A10-STMS differs from a full MPC, formal control-barrier-function controller, or flight controller in three respects. First, it is not an online optimal control design. Second, it does not claim a formal invariance theorem for the safe set. Third, it uses CVaR as a diagnostic adverse-tail metric rather than as a stochastic optimality proof. Its intended role is a closed intermediate surrogate for locating feasibility regimes, identifying dominant bottlenecks, and separating controller-limited from resource-limited failures.

The core contribution is therefore not a hardware design. It is a claim-boundary-controlled numerical audit showing that:

- under insufficient thermal resources, A10-Hybrid behavior is damage-limiting but not mission-feasible;
- after sufficient radiator and transport resources are introduced, an A10 separated-barrier controller can become mission-feasible in manageable stress families;
- valve routing, storage buffering, and load shaping each contribute nontrivially;
- delayed observations and actuator degradation create distinct bottlenecks that require explicit redesign;
- a final authority-aware A10-v4-safe controller preserves feasibility under baseline, delayed-observation, moderate actuator-degradation, and out-of-sample stress windows in the tested reduced surrogate;
- harsh combined stress remains outside the solved regime.

This scope-controlled positioning is essential. The numerical results below do not demonstrate thermal-vacuum validity, spacecraft readiness, or superiority over existing spacecraft thermal-control architectures. They demonstrate a reduced-surrogate feasibility separation.

2 A10-STMS Design Principle

A10-STMS inherits the general A10 engineering principle of using a structured prior to restrict control search to a mission-relevant, robustness-oriented sector rather than optimizing a free-form actuation policy. In earlier A10 engineering applications, the same conceptual pattern was used to separate mission variables from diagnostic variables, to emphasize safe-set survival rather than nominal best-case performance, and to treat resource insufficiency as an informative negative result rather than a numerical failure [8–10].

For thermal systems, this design principle becomes:

> A spaceborne industrial cooling controller should preserve mission feasibility within a multidimensional thermal safe set, not merely minimize one temperature signal.

The control problem is therefore formulated as a viability problem,

$$x(t) \in \mathcal{S}_{\mathrm{th}} \quad \forall t \in [0, T], \qquad Y(T) \geq Y_{\min}, \tag{1}$$

where $x(t)$ is the reduced thermal state, $\mathcal{S}_{\mathrm{th}}$ is a multidimensional thermal safe set, and $Y(T)$ is the final mission-throughput variable. This differs from scalar thermal optimization such as minimizing $\max_t T_c(t)$ because it treats thermal safety, power, storage, gradient risk, and mission throughput as jointly binding constraints.

3 Reduced Thermal Surrogate

3.1 State, safe set, and mission condition

The reduced state is

$$x(t) = (T_c, T_p, T_r, E_s, P_m, G, Y), \tag{2}$$

where T_c is core temperature, T_p is cold-plate temperature, T_r is radiator temperature, E_s is storage margin, P_m is power margin, G is thermal-gradient risk, and Y is accumulated mission throughput.

Table 1: Reduced state variables in the A10-STMS surrogate.

Symbol	Interpretation	Safety role
T_c	Core equipment temperature	Primary electronics / payload thermal limit
T_p	Cold-plate temperature	Heat-transport interface limit
T_r	Radiator temperature	Radiative saturation / rejection limit
E_s	Storage margin	Thermal buffer reserve proxy
P_m	Power margin	Avoids cooling effort consuming mission power
G	Thermal-gradient risk	Structural, optical, and local stress proxy
Y	Mission throughput	Useful data, compute, illumination, processing, or production

The safe set is

$$\mathcal{S}_{\mathrm{th}} = \left\{ \begin{array}{lll} T_c \leq T_c^{\max}, & T_p \leq T_p^{\max}, & T_r \leq T_r^{\max}, \\ E_s \geq E_s^{\min}, & P_m \geq P_m^{\min}, & G \leq G^{\max} \end{array} \right\}. \tag{3}$$

A run is counted as failed if at least one time step violates Eq. (3), if the numerical state becomes non-finite, or if the final mission condition $Y(T) < Y_{\min}$ holds. The reported *failure rate* is therefore a binary run-level statistic, whereas the *unsafe fraction* is the fraction of time steps within a run for which $x(t) \notin \mathcal{S}_{\mathrm{th}}$. This distinction is important: a run can have a small unsafe fraction but still count as failed.

3.2 Control variables

The control vector is

$$u(t) = (u_{\mathrm{pump}}, u_{\mathrm{valve}}, u_{\mathrm{rad}}, u_{\mathrm{store}}, u_{\mathrm{load}}), \tag{4}$$

where the five channels respectively represent heat-transport pumping, thermal-path routing, radiator/emissivity authority, storage coupling, and mission-load shaping. The essential A10-STMS hypothesis is that these channels should not be collapsed into a single "cooling effort" scalar. Power-like actuation, routing-like steering, storage use, radiative rejection, and demand-side shaping are assigned separated risk gates.

3.3 Dynamics

The nondimensional ODE surrogate is

$$\dot{T}_c = \frac{1}{C_c}\left(Q_{\text{gen}}(t, u_{\text{load}}) - Q_{cp}\right) + \xi_c, \tag{5}$$

$$\dot{T}_p = \frac{1}{C_p}\left(Q_{cp} - Q_{pr} - Q_{ps}\right) + \xi_p, \tag{6}$$

$$\dot{T}_r = \frac{1}{C_r}\left(Q_{pr} - Q_{\text{rad}} + Q_{\text{sun}}(t)\right) + \xi_r, \tag{7}$$

$$\dot{E}_s = -k_E Q_{ps} + k_{E,r} R_{\text{rec}}(t, u_{\text{rad}})(1 - E_s), \tag{8}$$

$$\dot{P}_m = k_P \left(P_{\text{in}}(t) - P_{\text{cool}}(u)\right), \tag{9}$$

$$\dot{G} = \frac{G_{\text{inst}}(x, u, \dot{u}) - G}{\tau_G}, \tag{10}$$

$$\dot{Y} = \frac{u_{\text{load}}}{T}. \tag{11}$$

Here Q_{cp} and Q_{pr} denote core-to-plate and plate-to-radiator transport, Q_{ps} denotes storage coupling, Q_{rad} is a Stefan–Boltzmann-like radiative rejection proxy, and Q_{sun} represents environmental solar or view-factor heating. The surrogate is deliberately nondimensional; its variables are not calibrated to a specific spacecraft, orbit, payload, material stack, or thermal-vacuum test.

Table 2: Parameter groups and audit treatment. The table records model closure rather than physical spacecraft calibration.

Parameter group	Role in the reduced surrogate	Treatment in audit
C_c, C_p, C_r	Thermal-capacitance proxies for core, cold plate, and radiator nodes	Fixed nondimensional constants
k_{cp}, k_{pr}	Core-to-plate and plate-to-radiator transport coefficients	Swept through transport scale
k_{rad}	Radiative rejection coefficient / effective radiator authority	Swept through radiator scale
k_{store}	Coupling to finite thermal storage buffer	Swept through storage scale
$T_i^{\max}$, $E_s^{\min}$, $P_m^{\min}$, $G^{\max}$	Hard safe-set thresholds	Fixed audit thresholds
$Y_{\min}$	Final mission-throughput threshold	Fixed at the validation target
$T, \Delta t$	Simulation horizon and time step	Fixed across comparisons
ξ_i and random multipliers	Monte Carlo perturbations of heat, transport, radiation, power, and noise	Sampled with declared seeds

3.4 Violation integral and CVaR diagnostic

The integrated violation functional is

$$V = \int_0^T \Big[(T_c - T_c^{\max})_+^2 + (T_p - T_p^{\max})_+^2 + (T_r - T_r^{\max})_+^2 \\ +(E_s^{\min} - E_s)_+^2 + (P_m^{\min} - P_m)_+^2 + (G - G^{\max})_+^2\Big]\, dt. \tag{12}$$

For a sampled population of violation outcomes V, the adverse-tail diagnostic is $\mathrm{CVaR}_{0.95}(V)$, defined as the mean of the worst 5% of violation-integral outcomes. CVaR is used here only as an engineering diagnostic of adverse-tail behavior, not as a formal probabilistic optimality certificate.

4 Separated-Barrier A10 Control

A10-STMS constructs risk gates from separated barriers,

$$B_T = \left[\frac{T_c - T_c^{\mathrm{warn}}}{T_c^{\max} - T_c^{\mathrm{warn}}}\right]_+, \tag{13}$$

$$B_S = \left[\frac{E_s^{\mathrm{warn}} - E_s}{E_s^{\mathrm{warn}} - E_s^{\min}}\right]_+, \tag{14}$$

$$B_P = \left[\frac{P_m^{\mathrm{warn}} - P_m}{P_m^{\mathrm{warn}} - P_m^{\min}}\right]_+, \tag{15}$$

$$B_G = \left[\frac{G - G^{\mathrm{warn}}}{G^{\max} - G^{\mathrm{warn}}}\right]_+. \tag{16}$$

These barriers are not claimed to be formal control barrier functions in the mathematical sense of [5]; rather, they are reduced surrogate gates inspired by safe-set control. The control law has the generic structure

$$u(t) = u_{\mathrm{nom}} + g_T(B_T)\Delta u_T + g_S(B_S)\Delta u_S + g_P(B_P)\Delta u_P + g_G(B_G)\Delta u_G, \tag{17}$$

where g_i are smooth clipped gate functions.

The validated sequence used four main A10 variants. A10-v1 was the original separated-barrier controller. A10-v2 raised baseline radiator/transport authority and preserved load more strongly. A10-v3 added delay prediction and conservative risk inflation. A10-v4 added authority-aware retuning, with two variants: `a10_v4_balanced` and `a10_v4_safe`. The final conservative controller is `a10_v4_safe`.

5 Validation and Reproducibility Protocol

The validation sequence was predeclared in phases. Each phase was allowed to fail, and negative results were treated as diagnostic information. Table 3 summarizes the locked outcomes. The primary high-resource candidate used in the final phases was

$$\text{radiator scale} = 24.0, \qquad \text{transport scale} = 12.0, \qquad \text{storage scale} = 4.0. \tag{18}$$

This high-resource requirement is part of the claim boundary. The paper does not claim feasibility in low-resource regimes.

Table 3: Numerical validation sequence and locked interpretation.

Phase	Purpose	Locked interpretation
Phase 1	Initial controller comparison under broad stresses	Damage-limiting but not mission-feasible; all modes failed by mission criteria
Phase 1T	Failure-reason triage	Dominant bottleneck was thermal / gradient; storage and power were not primary
Phase 1R	Resource-frontier sweep	Still not mission-feasible; resource frontier required
Phase 1R2	Extended resource sweep and A10-v2	Provisional high-resource mission-feasible separation
Phase 1R2C	Higher-sample confirmation	Confirmed high-resource separation with $n = 64$ per stress/mode/candidate
Phase 2	Ablation, OOS, delay, degradation	Baseline and ablation passed; delay/degradation remained weak
Phase 2D	Delay/degradation redesign	Delay improved strongly; actuator authority remained near boundary
Phase 2E	Authority-aware retuning	`PASS-PHASE2E-DELAY-DEGRADE-ROBUSTNESS`

Table 4: Computational audit counts for the final confirmation and robustness stages.

Stage	Family / condition	Samples represented	Interpretation
Phase 1R2C	Each stress/mode/candidate	$n = 64$	High-sample confirmation of the high-resource corner.
Phase 2E manageable	7 stress windows $\times$ 32	224 per mode/condition	Final manageable-stress robustness audit.
Phase 2E OOS	5 OOS windows $\times$ 32	160 per mode/condition	Out-of-sample stress audit.
Phase 2E frontier	1 frontier window $\times$ 32	32 per mode/condition	Damage-limiting boundary audit.
Phase 2E harsh	1 harsh window $\times$ 32	32 per mode/condition	Explicit negative-boundary audit.

The numerical scripts used for the staged audit were named according to their phase: `a10_stms_phase1.py`, `a10_stms_phase1R2_extended_frontier.py`, `a10_stms_phase1R2C_confirm.py`, `a10_stms_phase2_ablation_oos.py`, `a10_stms_phase2D_delay_degrade_redesign.py`, `a10_stms_phase2E_authority_retune.py`, and `a10_stms_phase3_synthesis.py`. The present manuscript reports the locked outputs of those scripts. Future archival release should pair this manuscript with the scripts, output CSV files, and figure-generation logs.

6 Phase 1: Damage Limiting but Not Feasibility

The initial Phase 1 experiment showed an important but incomplete result. A10-Hybrid minimized the worst-tail violation integral, but all modes had failure rate equal to one because the model was too thermally severe and mission throughput was sacrificed. Figure 1 shows the Phase 1 CVaR comparison.

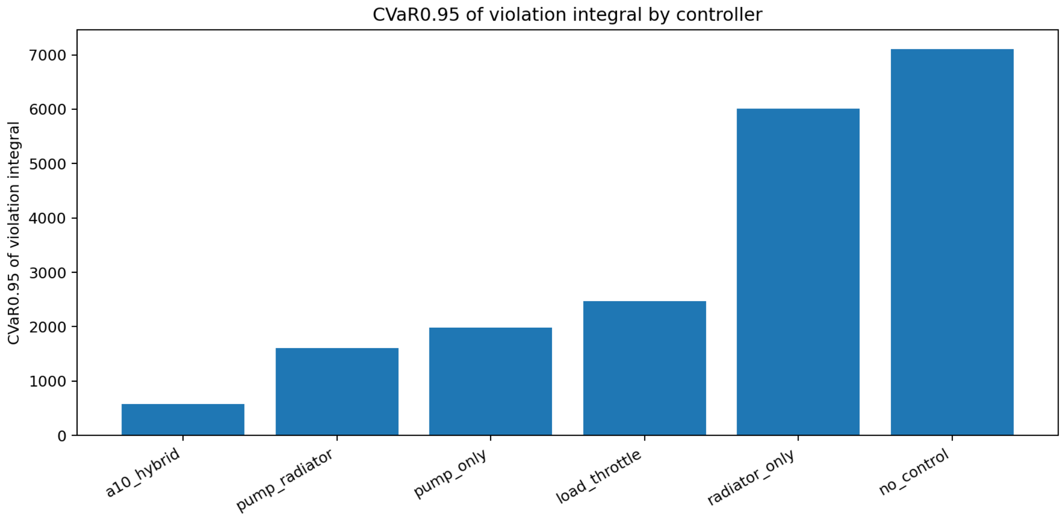


Figure 1: Phase 1 damage-limiting result. A10-Hybrid strongly reduced the tail violation integral relative to Pump+Radiator and uncontrolled baselines, but this did not constitute mission feasibility because all controllers still failed the full safety-throughput condition.

The correct Phase 1 interpretation is therefore not “cooling success,” but “damage-limiting under resource insufficiency.” This motivated explicit resource-frontier analysis.

7 Phase 1R2C: Resource-Frontier Discovery and Confirmation

The initial resource-frontier sweep remained insufficient. Phase 1R2 extended the grid and introduced A10-Hybrid-v2. The first clean mission-feasible separation appeared only at the high-resource corner, especially near radiator scale 24 and transport scale 12. Figure 2 illustrates the extended resource-frontier classification.

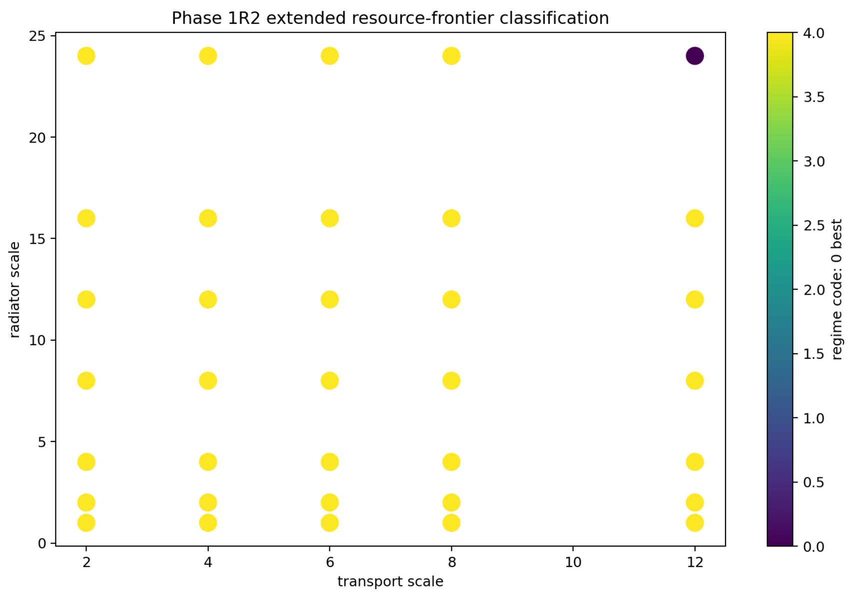
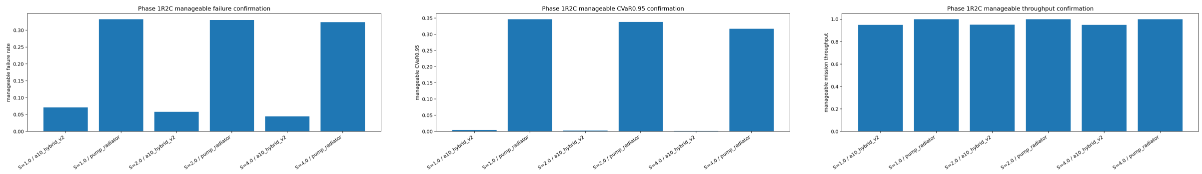


Figure 2: Phase 1R2 extended resource-frontier classification. Mission-feasible separation appears only in the high-resource regime, supporting a resource-frontier rather than universal-controller interpretation.

Phase 1R2C confirmed the high-resource result with higher Monte Carlo sampling. Figure 3 compares manageable failure rate, CVaR, and mission throughput for A10-Hybrid-v2 versus Pump+Radiator across storage scales.



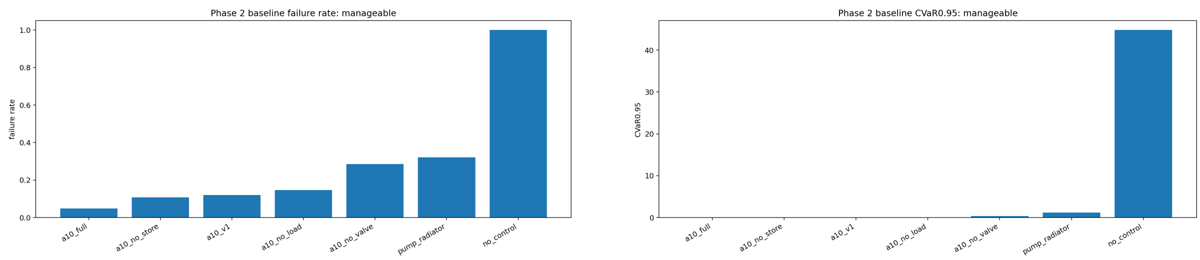
(a) Failure rate **(b)** $\mathrm{CVaR}_{0.95}$ **(c)** Throughput

Figure 3: Phase 1R2C high-sample confirmation. A10-Hybrid-v2 preserves lower failure and lower tail risk than Pump+Radiator while maintaining high mission throughput in the high-resource regime.

The Phase 1R2C representative high-resource outcome was failure rate 0.0714, $\mathrm{CVaR}_{0.95} = 0.00446$, unsafe fraction 0.00932, and mean throughput 0.949 for the confirmed A10-Hybrid-v2 candidate. The result should be read as a high-resource feasibility point, not a universal controller guarantee.

8 Phase 2: Ablation and Out-of-Sample Stress

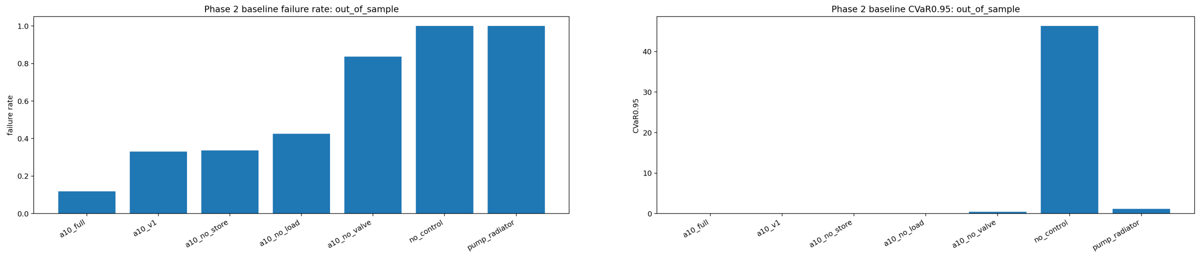
Phase 2 focused on the high-resource candidate with storage scale 4.0. The baseline manageable-stress comparison is shown in Figure 4. A10-Full, corresponding to the validated A10-v2 architecture before later delay/degradation redesign, achieved low manageable failure and near-zero manageable CVaR relative to baselines and ablations.



(a) Manageable failure rate **(b)** Manageable $\mathrm{CVaR}_{0.95}$

Figure 4: Phase 2 baseline manageable-stress comparison. A10-Full is separated from Pump+Radiator and from ablated A10 variants. The ablations indicate that valve routing, storage buffering, and load shaping are all nontrivial contributors.

Out-of-sample stress was then used to test whether the result was specific to the stress windows used in controller construction. Figure 5 shows that A10-Full retains a substantial separation from Pump+Radiator in both failure rate and CVaR.



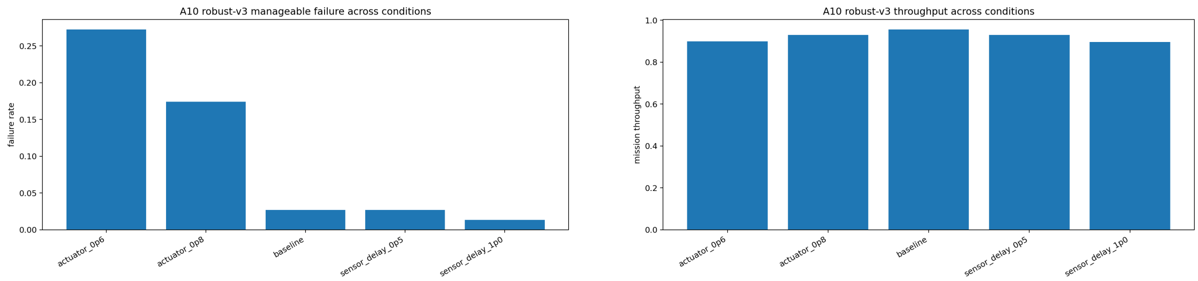
(a) OOS failure rate **(b)** OOS $\mathrm{CVaR}_{0.95}$

Figure 5: Phase 2 out-of-sample behavior. A10-Full retains lower failure and lower tail-risk than Pump+Radiator under stress windows not used to construct the initial controller.

However, Phase 2 also exposed a serious weakness: delayed sensing and actuator degradation were not adequately handled by the raw A10-Full controller. This triggered Phase 2D.

9 Phase 2D and Phase 2E: Delay and Actuator Robustness

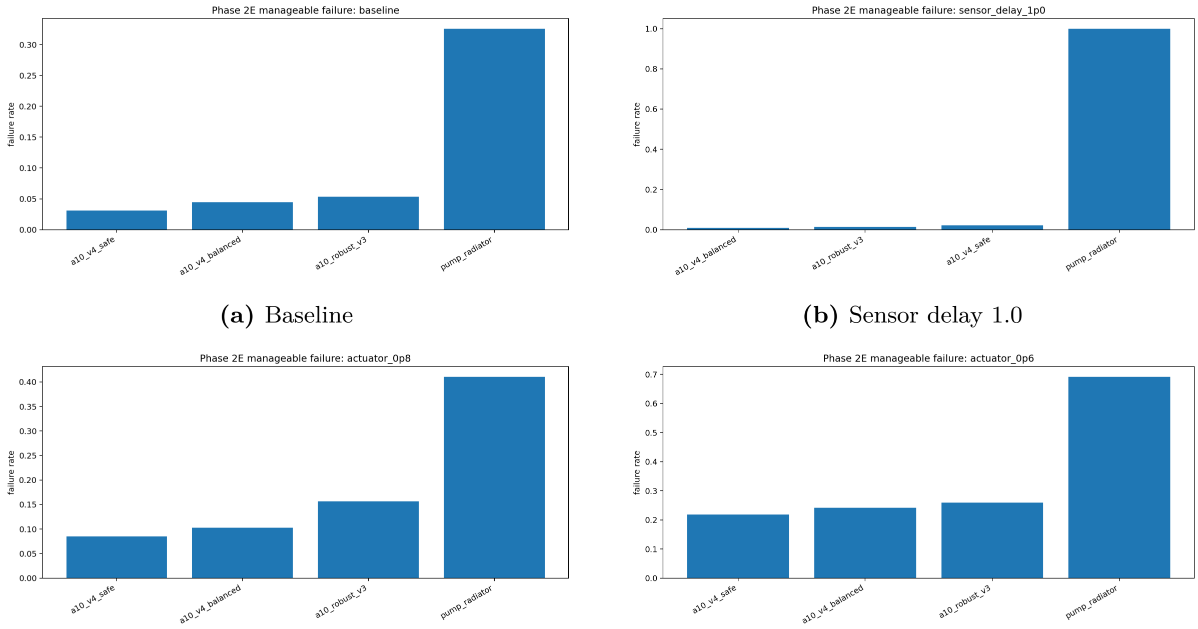
Phase 2D introduced a simple model-based delay predictor and an A10 robust-v3 controller. The delay bottleneck was substantially reduced, while throughput remained above the mission threshold. Figure 6 summarizes robust-v3 behavior across delay and degradation conditions.



(a) Failure across conditions **(b)** Throughput across conditions

Figure 6: Phase 2D delay/degradation redesign. A10 robust-v3 strongly improves delayed-sensing performance while preserving mission throughput, but actuator degradation remains a near-boundary bottleneck.

Phase 2E then introduced authority-aware A10-v4 variants. The conservative A10-v4-safe controller is the final preferred controller. Figure 7 shows manageable failure-rate comparisons under baseline, one-unit sensor delay, 0.8 actuator authority, and 0.6 actuator authority.



(a) Baseline

(b) Sensor delay 1.0

(c) Actuator authority 0.8

(d) Actuator authority 0.6

Figure 7: Phase 2E authority-aware retuning. A10-v4-safe passes baseline, delayed observation, and moderate actuator degradation. The 0.6 actuator-authority condition remains a soft-pass / residual-risk regime rather than a full mission-feasible guarantee.

Out-of-sample tail-risk separation is preserved after Phase 2E, as shown in Figure 8. Pump+Radiator retains much larger tail risk than the A10-v4 family.

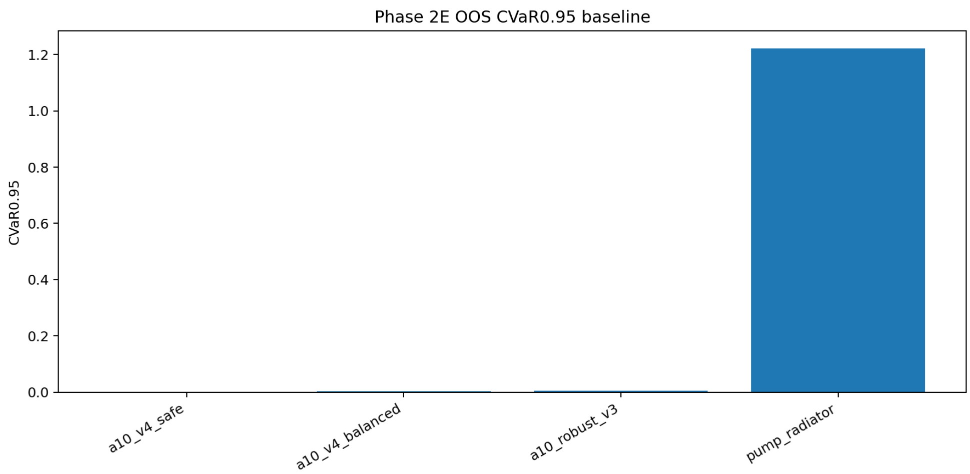


Figure 8: Phase 2E out-of-sample $\mathrm{CVaR}_{0.95}$ comparison. The A10-v4 controllers retain strong tail-risk separation from Pump+Radiator.

10 Final Metrics and Claim Boundary

The final representative metrics are summarized in Tables 5 and 6. Values are nondimensional reduced-surrogate metrics and should not be interpreted as physical spacecraft performance levels.

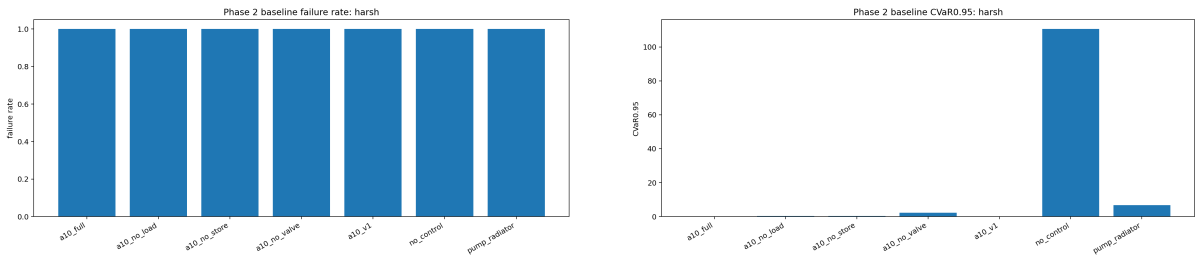
Table 5: Final Phase 2E manageable-stress metrics. These are numerical audit metrics, not flight-test data.

Condition	Controller	n	Failure	$\text{CVaR}_{0.95}$	Unsafe frac.	Throughput	T_c p95
Baseline	A10-v4-safe	224	0.03125	4.6976e-05	0.003051	0.964707	0.991673
Baseline	Pump+Radiator	224	0.325893	0.943606	0.099892	1.000000	1.364300
Delay 0.5	A10-v4-safe	224	0.022321	1.1430e-06	0.001079	0.928990	0.977225
Delay 0.5	Pump+Radiator	224	0.727679	1.211720	0.109479	1.000000	1.413250
Delay 1.0	A10-v4-balanced	224	0.008929	4.4956e-06	0.000911	0.898742	0.951084
Delay 1.0	Pump+Radiator	224	1.000000	1.136450	0.151283	1.000000	1.408560
Actuator 0.8	A10-v4-safe	224	0.084821	0.008999	0.010182	0.895611	1.028617
Actuator 0.8	Pump+Radiator	224	0.410714	1.579370	0.197187	1.000000	1.518670
Actuator 0.6	A10-v4-safe	224	0.218750	0.060507	0.025194	0.820892	1.094786
Actuator 0.6	Pump+Radiator	224	0.691964	2.436140	0.496722	1.000000	1.654540

Table 6: Out-of-sample, frontier, and harsh baseline metrics for final A10-v4-safe compared with Pump+Radiator.

Family	Controller	n	Failure	$\text{CVaR}_{0.95}$	Unsafe frac.	Throughput	T_c p95
Out-of-sample	A10-v4-safe	160	0.068750	0.001404	0.006667	0.943630	1.010980
Out-of-sample	Pump+Radiator	160	0.993750	1.223330	0.396042	1.000000	1.411170
Frontier	A10-v4-safe	32	0.343750	0.003098	0.034609	0.931709	1.034860
Frontier	Pump+Radiator	32	1.000000	1.389970	0.552839	1.000000	1.456240
Harsh	A10-v4-safe	32	0.968750	0.189089	0.164453	0.875229	1.209230
Harsh	Pump+Radiator	32	1.000000	4.413550	0.905052	1.000000	1.696570

The harsh combined stress case remains outside the solved regime. Figure 9 shows that all tested controllers fail by failure-rate criteria under harsh stress, even when A10 reduces CVaR relative to baselines. This is not a weakness to hide; it is an important claim boundary.



(a) Harsh failure rate

(b) Harsh $\text{CVaR}_{0.95}$

Figure 9: Harsh combined stress. A10 reduces tail damage in the reduced surrogate, but harsh stress is not mission-feasible and is excluded from the success claim.

The final defensible statement is therefore:

> A10-STMS is mission-feasible only in the tested high-resource regime and for manageable, delayed-observation, moderate actuator-degradation, and out-of-sample stress families. Harsh combined stress is resource- or architecture-limited in the current surrogate.

11 Discussion

11.1 Why the separated-barrier architecture matters

The ablation results suggest that A10-STMS is not merely a stronger pump or a more aggressive radiator. Removing valve routing, storage coupling, load shaping, or the v2/v4 risk-structured logic degrades the result. This supports the interpretation that the advantage comes from separated risk handling:

- thermal barriers trigger transport and radiator authority;
- storage barriers prevent overuse of finite thermal buffers;
- power barriers prevent cooling from destroying mission power margin;
- gradient barriers avoid excessive thermal-routing aggressiveness;
- load shaping preserves mission viability by reducing demand only when the thermal state requires it.

The distinction between A10-v4-safe and A10-v4-balanced is also useful. A10-v4-safe gives the cleanest conservative robustness, while A10-v4-balanced sometimes preserves slightly better throughput under delayed observations. For a real engineering program, the choice would depend on mission criticality, payload degradation tolerance, and available authority margins.

11.2 Industrial relevance

The relevant applications are not low-power spacecraft subsystems whose thermal loads are already comfortably handled by passive design. A10-STMS is aimed at high-duty-cycle or bursty spaceborne industrial systems where thermal, power, and mission variables are strongly coupled. Examples include orbital compute payloads, high-throughput communications, laser communication terminals, electric-propulsion power electronics, high-power sensors, and in-space manufacturing nodes.

The practical engineering value of the framework is diagnostic. It can answer whether failure is due to controller tuning or insufficient radiator / transport resources, whether load shaping is necessary or Pump+Radiator is sufficient, whether delayed sensing is a dominant bottleneck, how much actuator degradation can be tolerated before mission infeasibility, and whether a candidate architecture is mission-feasible, damage-limiting, near-frontier, or architecture-limited.

12 Limitations

The limitations are central to the paper and should remain visible.

1. The model is nondimensional and reduced. It is not calibrated to a spacecraft, payload, orbit, material stack, or thermal-vacuum experiment.
2. Radiative heat rejection is represented by a simplified fourth-power proxy and does not include detailed view factors, orbital beta-angle histories, multi-node geometry, or spacecraft attitude dynamics.
3. Storage is represented by a scalar margin rather than a phase-change material model or detailed heat-capacity network.
4. Power margin is reduced to a scalar reserve and does not model full spacecraft power-system dynamics.
5. The delay predictor is intentionally simple and should not be confused with a flight-ready estimator.

6. CVaR is used as a diagnostic tail-risk metric, not as a formal guarantee of stochastic optimality.
7. Harsh combined stress remains unsolved and is explicitly excluded from the mission-feasible claim.
8. The current figure and table package reports locked outputs; a separate reproducibility archive should include the exact scripts, seeds, and output CSV files.

13 Conclusion

A10-STMS provides a reduced, mission-variable-preserving thermal-control surrogate for space-borne industrial systems. The numerical sequence shows a progression from negative results to resource-frontier discovery to robust high-resource feasibility. The initial controller was only damage-limiting; after radiator and transport resources were increased, A10-Hybrid-v2 achieved mission-feasible separation; after ablation and out-of-sample tests, delayed observations and actuator degradation were identified as bottlenecks; and after authority-aware retuning, A10-v4-safe produced the final locked status:

`PASS-PHASE2E-DELAY-DEGRADE-ROBUSTNESS.`

The contribution is not a new thermal law or a spacecraft-ready cooling system. It is a closed intermediate surrogate and feasibility-diagnostic architecture showing that separated-barrier hybrid control can preserve mission variables under certain high-resource thermal regimes better than simple Pump+Radiator baselines. The result justifies a next-stage model with calibrated thermal nodes, realistic orbit/environment forcing, detailed radiator/view-factor geometry, power-system constraints, and hardware-relevant actuator models.

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Conflict of Interest

The author declares no institutional, financial, or commercial conflict of interest. This manuscript is an independent theoretical and numerical surrogate study.

Data and Code Availability

The manuscript package contains the LaTeX source and the figures used in the manuscript. The numerical audit was generated by phase-specific scripts named in Section 5. For formal reuse, those scripts, seeds, and CSV outputs should be archived together as a supplementary reproducibility package.

References

[1] J. Meseguer, I. Pérez-Grande, and A. Sanz-Andrés, *Spacecraft Thermal Control*, Woodhead Publishing, 2012. doi:10.1533/9780857096081.

[2] D. G. Gilmore, ed., *Spacecraft Thermal Control Handbook, Volume I: Fundamental Technologies*, 2nd ed., Aerospace Press, 2002.

[3] R. D. Karam, *Satellite Thermal Control for Systems Engineers*, Progress in Astronautics and Aeronautics, Vol. 181, AIAA, 1998.

[4] R. T. Rockafellar and S. Uryasev, "Optimization of Conditional Value-at-Risk," *The Journal of Risk*, Vol. 2, No. 3, pp. 21–42, 2000. doi:10.21314/JOR.2000.038.

[5] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control Barrier Function Based Quadratic Programs for Safety Critical Systems," *IEEE Transactions on Automatic Control*, Vol. 62, No. 8, pp. 3861–3876, 2017. doi:10.1109/TAC.2016.2638961.

[6] S. J. Qin and T. A. Badgwell, "A Survey of Industrial Model Predictive Control Technology," *Control Engineering Practice*, Vol. 11, No. 7, pp. 733–764, 2003. doi:10.1016/S0967-0661(02)00186-7.

[7] O. J. M. Smith, "Closer Control of Loops with Dead Time," *Chemical Engineering Progress*, Vol. 53, No. 5, pp. 217–219, 1957.

[8] K. Yoshimura, "A10-PTF: Mission-Variable-Preserving Structured Priors under Uncertainty," independent manuscript, 2026.

[9] K. Yoshimura, "UAPST-II: Mission Feasibility, Delay-Critical Hybridity, and Identifiability in a Reduced Boundary-MHD Surrogate," independent manuscript, 2026.

[10] K. Yoshimura, "A10-RPT: Nonlinear Safety Envelope Audit and Resource-Frontier Diagnosis," independent manuscript, 2026.